CHM 120

Limiting Reactants

Final Report

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Lesson	Limiting Reactants
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Test Your Knowledge

Nitrogen and hydrogen react to form ammonia:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

Identify the limiting reactant (hydrogen or nitrogen) in each of the following combinations of starting chemicals.

Limiting Reactant: Hydrogen	Limiting Reactant: Nitrogen
1	2
1.0 mole of nitrogen and 1.0 mole of hydrogen	0.75 moles of nitrogen and 2.5 moles of hydrogen
	1.0 mole of nitrogen and 4.0 moles of hydrogen
	10.0 grams of nitrogen and 10.0 grams of hydrogen

Label each type of reaction.

$Al_2(SO_4)_3\,(aq) + 6NaOH\,(aq) \rightarrow 2Al(OH)_3\,(s) + 3Na_2SO_4\,(aq)$

1 Double displacement

$2KClO_3\,(s) \rightarrow 2KCl\,(s) + 3O_2\,(g)$

2 Decomposition

$4Fe\,(s) + 3O_2\,(g) \rightarrow 2Fe_2O_3\,(s)$

3 Combination

$Cl_2\,(g) + 2NaI\,(aq) \rightarrow 2NaCl\,(aq) + I_2\,(aq)$

4 Single displacement

Move the terms to complete each statement.

A 1 [Chemical equation] describes a reaction that occurs between two or more chemical substances.

A chemical substance that participates and undergoes change in a chemical reaction is called a 2 [Reactant] .

3 [Stoichiometry] can be used to determine how much of each reactant is needed to produce a specific amount of each product.

The reactant that is consumed first is referred to as the 4 [Limiting reactant] .

The 5 [Law of conservation of mass] states that the total mass, in a closed system, does not change as the result of reactions between its parts.

Exploration

The substances that interact in a reaction are called reactants, while the substances produced from the reaction are called products.

- (•) True
- ( ) False

The proportion of reactants to products can be calculated using a balanced chemical equation and stoichiometry.

- (•) True
- ( ) False

A _____ reaction occurs when an uncombined element displaces an element in a compound.

- decomposition
- combination
- single displacement
- double displacement

In a chemical reaction, the reactant that is consumed first is referred to as the _____.

- stoichiometry
- smaller reactant
- limiting reactant
- balanced equation
- limiting product

In the chemical equation $2Ca + O_2 \rightarrow 2CaO$, if calcium is the limiting reactant, then _____ will be in excess.

- Ca
- O_2
- CaO
- no substances

The law of conservation of mass states that _____.

- the total mass, in an open system, does not change as the result of reactions between its parts
- the total mass, in a closed system, does not change as the result of reactions between its parts
- particles at rest tend to stay at rest
- the total energy of an isolated system cannot change

Exercise 1

From your calculations, which reaction experiment had closest to stoichiometric quantities? How many moles of $NaHCO_3$ and $HC_2H_3O_2$ were present in this reaction?

B I U 0 Word(s)

The chemical reaction you investigated is a two-step reaction. What type of reaction occurs in each step? How did you determine your answer?

B I U 38 Word(s)

First Step: Double Displacement --> Two elements forming a compound along with two other elements that are also a compound, change to create two new compounds.

Second Step: Decomposition as the one compound breaks down into it's components.

As the reaction was theoretically performed in a closed system, there should be no change in mass before and after the reaction. What are possible sources of error in your experiment reactions?

B I U 55 Word(s)

Some of the mass of the sodium bicarbonate may not have fallen into the balloon for whatever reason (e.g. it stuck to somewhere on the inside of the balloon), some of the vinegar may not have been poured out all of the way resulting in the next tube possibly having more than the last tube.

What is the relationship between the limiting reactant and theoretical yield of CO_2?

B I U 23 Word(s)

The relationship is that the theoretical yield displays what would happen ideally if all of the limiting reactant is used in some reaction

After the reactions occurred, did the largest balloon size match the largest theoretical yield you calculated in Data Table 4? Use your experimental results to explain your answer. If not, what sources of error do you think caused the discrepancy?

B *I* U 27 Word(s)

I believe that the largest balloon was the first balloon (red) and it matched what I believe to be the calculated theoretical yield of CO_2.

Data Table 1: Conservation of Mass – Initial Mass

Reaction #	Mass of test tube and 5.0% $HC_2H_3O_2$ (g) (A)
1	17.6g
2	17.5g
3	18.3g
4	18.1g

Reaction #	Mass of $NaHCO_3$ (g) (B)
1	0.10
2	0.20
3	0.35
4	0.50

Reaction #	Mass of balloon and $NaHCO_3$ (g) (C)
1	1.37g
2	1.75g
3	2.08g
4	2.13g

Reaction #	Total mass before reaction (g) (D = A + C)
1	19.0g
2	19.3g
3	20.4g
4	20.2g

Data Table 2: Moles of HC2H3O2

Reaction #	Volume of 5.0% Vinegar (mL)	Mass of $HC_2H_3O_2$ (g)	Moles of $HC_2H_3O_2$
1	5.0	0.25	.004mol
2	5.0	0.25	.004mol
3	5.0	0.25	.004mol
4	5.0	0.25	.004mol

Data Table 3: Moles of NaHCO3

Reaction #	Mass of $NaHCO_3$ (g)	Moles of $NaHCO_3$
1	.10g	.001mol
2	.20	.002mol
3	.35	.004mol
4	.50	.006mol

Data Table 4: Theoretical Yield of CO2

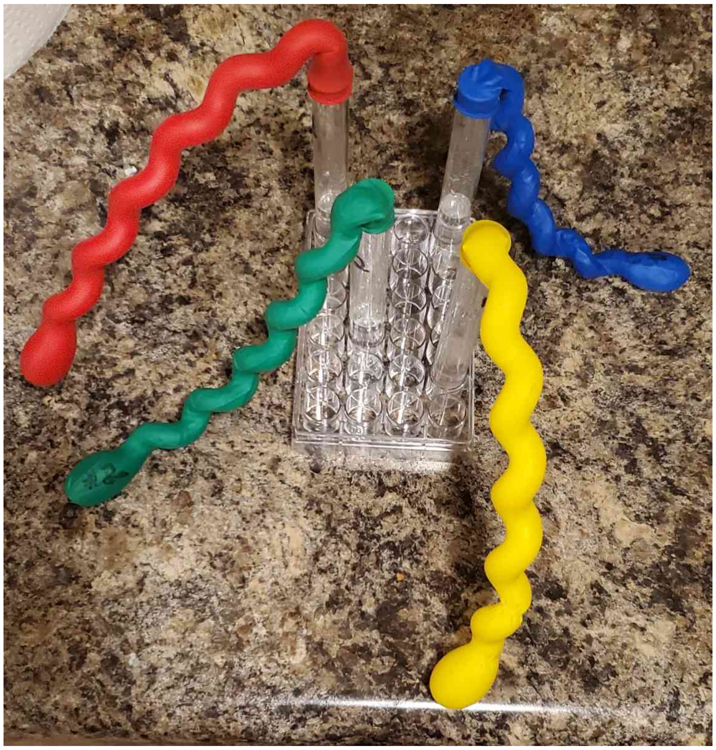
Reaction #	Limiting Reactant	Theoretical yield of CO_2 (moles)	Theoretical yield of CO_2 (g)
1	$NaHCO_3$	$1mol$	$.90g$
2	$NaHCO_3$	$1mol$	$.80$
3	$NaHCO_3$	$1mol$	$.65g$
4	$NaHCO_3$	$1mol$	$.50g$

Panel 1: Hypothesis

B *I* U 33 Word(s)

I think that baloon #4 will expand the most. It has the most sodium bicarbonate which offers the vinegar more of something to react with, ultimately ending it the release of more CO_2

Photo 1: Reaction Results



Data Table 5: Conservation of Mass

Reaction #	Total Mass after Reaction (g) (E)	Mass Difference (g) (F = D - E)
1	19.1g	.1g
2	19.1g	.2g
3	20.2g	.2g
4	20.0g	.2g

Competency Review

In the chemical equation $4\mathrm{Fe} + 3\mathrm{O}_2 \rightarrow 2\mathrm{Fe}_2\mathrm{O}_3$, 4Fe is considered a _____.

- (selected) reactant
- product
- compound

Stoichiometry can be applied to a chemical equation to calculate _____.

- the half-life of an element
- significant digits based on SI units
- activation energy in relation to chemical reactivity
- (selected) the proportion of reactants and products

When chlorine reacts with sodium iodide to form sodium chloride and iodine, a _____ reaction takes place.

- combination
- decomposition
- (selected) single displacement
- double displacement

Consider a closed system in which aluminum sulfate and sodium hydroxide react to form aluminum hydroxide and sodium sulfate. If aluminum sulfate is the limiting reactant, what other information can be concluded about the substances?

- the reaction rate will decrease
- the sodium hydroxide will be found in excess
- the reverse reaction will be favored
- a single displacement reaction will proceed

The law of conservation of mass states that the total mass, in a closed system, does not change as the result of reactions between its parts.

- True
- False

The law of conservation of mass states that the total mass in a closed chemical system does not change. In a student experiment, initial mass might differ from final mass due to experimental error.

- True
- False

$$\mathrm{NaHCO_3(s)\ +\ HC_2H_3O_2(aq)\ \rightarrow\ C_2H_3O_2Na(aq)\ + H_2CO_3(aq)}$$
$$\mathrm{H_2CO_3(aq) \rightarrow\ H_2O(l)\ + CO_2(g)}$$

If 0.00416 mols of $HC_2H_3O_2$ are reacted with 0.00595 mols of $NaHCO_3$, _____ is/are considered a limiting reactant.

- $HC_2H_3O_2$ and $NaHCO_3$
- $NaHCO_3$
- $HC_2H_3O_2$
- the products
- no chemicals

Percent yield can be calculated by dividing actual yield by theoretical yield and multiplying by 100.

- (selected) True
- False

Extension Questions

$2Al(s) + 3H_2SO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$

a. Determine the volume (mL) of 15.0 M sulfuric acid needed to react with 45.0 g of aluminum to produce aluminum sulfate.

b. Determine the % yield if 112 g of aluminum sulfate is produced under the above conditions.

.2mL needed to react